Cross-Border AI Architecture Patterns

Three Patterns, and Why the Fourth Is Not a Choice

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Canonical name. This framework is registry entry **REG-09** of the Defensible AI Framework Registry [1], which fixes its canonical name, its relationships and its position. **It is the only subordinate entry occupying no MESA altitude**, sitting instead under the Boundary Invariant applied to residency, a member of the data boundary class [2, 2.5; 4, 2.1], which the registry calls the data residency boundary class at 4.9.7. The reason is recorded at 6.5 of the registry.

Abstract

An institution told that data may not leave a jurisdiction usually treats the statement as an obstacle to a deployment. It is not an obstacle. It is the first fact about the deployment's architecture, and treating it as an obstacle is how an institution arrives at a topology chosen for cost and then argues that the clauses permit it.

This specification inverts that order. It is the **worked instance of the Boundary Invariant** [2] applied to the data residency boundary class: the clause is the boundary, the topology is the optimization, and the patterns are optimizations under a clause that may not be crossed. Three patterns are enumerated at the level of a single workload class. **Sovereign Silo** where no data may leave. **Federated** where named derived artifacts may move and the underlying data may not. **Regional Hub** where transfer is permitted to a destination every participating jurisdiction accepts. **Hybrid** is a different kind of thing: the composition that appears when workload classes in one estate resolve differently, reported across an estate rather than selected for a class.

This specification takes no residency rule as its own. Its single input is the **residency rule set** produced by the AI Data Governance Framework [4], consumed through a schema published there and cited here rather than restated. A selection references clause identifiers and **MUST NOT introduce a clause**, which prevents an architect from supplying, at the moment of selection, the permission the selection needs.

Section 4 is the decision instrument the registry recorded as missing from this entry, a limitation registry version 2.0 records as closed by that section [1]. It is deterministic. Two architects given one rule set and one workload classification reach the same assignment, or the rule set is defective in a way the instrument names. One blocking state, `no-pattern-resolves`, is the framework's falsification condition expressed as a recordable value rather than as a paragraph.

No institution unconnected to the author has been observed operating the selection as specified here. Section 8 states that, states that the registry, from its version 2.0, records this framework's authority as granted by CADRE, the AI Governance Operating Model (REG-07), and states what would falsify it. It is a specification, not a certification scheme, and no conformity assessment body operates against it.

Keywords: AI governance; data residency; cross-border AI; deployment architecture; boundary class; evidence architecture; sovereign AI; federated learning; regional hub; governed production AI.

Notation

This specification uses **MUST**, **MUST NOT**, **SHOULD**, **SHOULD NOT** and **MAY** as the normative verbal forms.

- **MUST** indicates a requirement. An implementation that does not meet it is not conformant with this specification.
- **SHOULD** indicates a recommendation. Departure requires a recorded rationale.
- **MAY** indicates a permission.

Normative terms here bind an institution's engineering and governance practice. They do not state, interpret or qualify any legal obligation, and this specification names no jurisdiction and states no transfer rule, for the reason at 2.3.

Defined terms

Residency clause, **disposition** and **residency rule set** carry the meanings fixed by the AI Data Governance Framework [4], which produces them. They are not redefined here.

Workload class. A set of AI workloads engaging the same set of residency clauses. **The workload class is the unit at which a pattern is selected**, for the reason at 2.4.

Selection. The act of assigning each workload class in scope a pattern or a blocking state, performed by the procedure at Section 4 and recorded in the `selection` object of the shared schema.

Routing evidence. The per-request record showing where a request was served and which clause set governed it. It is the artifact that distinguishes a compliance position from a topology choice.

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Normative status of sections

Sections 2 through 6 are normative. Sections 1, 7, 8 and 9 are informative, except where they state a requirement using **MUST**.

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1. Introduction

1.1 The problem

Two failures recur, and they are one failure in opposite directions.

The topology is chosen and the clauses are then found to permit it. Cost, latency and operating convenience select a shape, and the residency question arrives afterwards as a review. A review conducted after the shape is chosen has one available finding, because the alternative is expensive.

Or the clause is treated as a veto and the deployment does not happen. An institution reading a clause forbidding movement concludes the workload cannot be served, when what the clause has done is remove two of three architectures and leave one standing.

Both follow from the same error: **treating residency as a constraint on a decision rather than as the input that determines it.**

Both are errors of order.

1.2 What this framework specifies

A deterministic procedure that turns a set of residency clauses into a topology assignment per workload class, with named blocking states where the clauses do not determine one.

A residency rule determines the architecture the deployment takes. It does not stop a deployment. It shapes one.

1.3 Contribution

1. **Three patterns enumerated at the level of a workload class**, each keyed to a clause disposition rather than to a description of an institution. Section 3.
2. **Hybrid stated as a composition rather than as a fourth alternative**, which keeps the enumeration of three falsifiable. 3.4.
3. **A decision instrument**, the artifact the registry recorded as missing from this entry until its version 2.0 recorded that limitation as closed [1], deterministic and with its blocking states specified. Section 4.
4. **A requirement that the framework record its own authority**, at 2.5, which the register's grant from CADRE (REG-07) leaves in force.

1.4 Scope

Institutions operating AI workloads whose data is reached by at least one residency clause in more than one jurisdiction. **A workload class engaging no clause is outside this framework's scope** and MUST be recorded as out of scope rather than assigned a pattern, per 4.4.

1.5 Non-goals

- **Not a statement of any residency, transfer or jurisdiction rule.** The rule set is an input, produced elsewhere and sourced by the institution. See 2.3.
- **Not an infrastructure or network specification.** The patterns govern where inference and data may sit. Procurement, network design and contractual arrangements are outside them.
- **Not a cost model.** No figure has been measured for the patterns as specified here, and none is stated.
- **Not a certification scheme.** Section 6 defines the only claim this specification supports, and it is self-declared.
- **Not an empirical claim.** Section 8 states what has not been tested.

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2. Conceptual Model (Normative)

2.1 A worked instance, not a framework of its own

This framework is the worked instance of the Boundary Invariant [2] applied to the data residency boundary class, and MUST NOT be read as proposing a law, a class or a taxonomy of its own.

The division it inherits is exact. **The clause is the boundary**: the optimizer may not cross it. **The topology is the optimization**: everything about where compute, storage and serving sit that the clauses do not fix. The three patterns are three optimizations under one uncrossable clause, which is why they are patterns rather than strategies.

The consequence is that a residency decision made here carries a BOE Declaration in the same shape as a gate record, a validation record and a screening attestation [1]. That is what makes routing evidence usable during an incident reconstruction. Available is not usable.

2.2 No altitude

This framework occupies no MESA altitude [5]. It is one of two registry entries in that position and the only subordinate one, the other being the entry that defines the altitudes rather than standing outside them [1].

The absence is a property recorded by the registry at 4.9.7 and 6.5, not a field left blank. An implementation MUST NOT assign this framework an altitude in order to fit an altitude-indexed control inventory, because doing so places a residency decision under a governance level that does not bind it and obscures the law that does.

2.3 The input is authored elsewhere

The single input to this framework is the residency rule set produced by the AI Data Governance Framework [4], expressed in the schema published with that specification as `residency-rule-set-schema-v1.0.json`. This document cites that schema and does not republish it.

A selection MUST reference clause identifiers present in the rule set and MUST NOT introduce a clause. This is the closure rule of the registry [1] applied to the artifact the two frameworks exchange, and it is the requirement that does the most work here. An architect who may write a clause at the moment of selection can supply exactly the permission the selection needs. The resulting document will be internally consistent. It will have proved nothing.

An architect finding a needed clause absent, unresolved or expired has found a defect in the rule set, and MUST return it to the producing framework and record a blocking state, per 4.4.

This specification names no jurisdiction and states no transfer rule. Jurisdiction-specific rules age, and a specification carrying them would age with them while continuing to read as authoritative. Every clause carries its own primary source and re-verification date under [4], and a clause relied on past that date is not a clause.

2.4 The workload class is the unit of selection

A pattern MUST be selected per workload class, and MUST NOT be selected per system or per estate.

Selecting per system produces an estate of unrelated topologies with no statement of what they have in common. Selecting per estate forces the most restrictive clause in the estate onto every workload in it. A whole institution then pays for its smallest sensitive data set. The class is the unit.

Two workloads belong to one class when they engage the same set of clauses. The class is derived from the rule set, not from an organizational chart. A class asserted without a clause set behind it is a grouping. It is not a workload class.

2.5 The framework MUST record the authority it operates under

From version 2.0, the registry's relationship register records an authority grant to this framework [1]. Row 10 carries it from CADRE, the AI Governance Operating Model (REG-07), which now grants authority to six frameworks rather than five. **Through it the decision rights CADRE assigns reach this framework: the authority to fix a deployment topology, held by the named enterprise architect.**

Version 1.0 of this specification recorded the absence of that grant as a real gap and did not work around it, because the registry's own asymmetry claim is that evidence cannot be produced without authority: a control that

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no altitude required is a control no one is accountable for [1]. This framework produces evidence, and the register now records the grant behind it.

The registry records that its grant leaves the requirement below unaffected [1]. **An institution operating this framework MUST record which body granted its named enterprise architect the authority to fix a deployment topology, in the same form it records other authority grants.** An institution that cannot name that body has found the gap in its own governance rather than in this document.

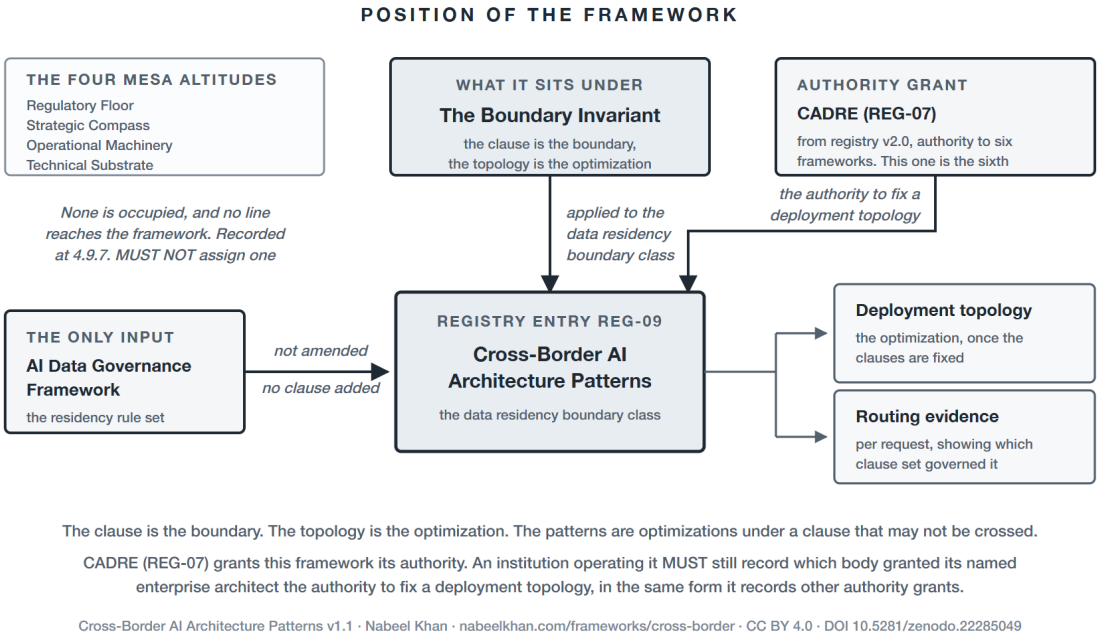


Figure 1. The framework's position. It sits under the Boundary Invariant rather than under a MESA altitude, taking the residency rule set from the AI Data Governance Framework as its only input and supplying the deployment topology and per-request routing evidence onward. The clause is the boundary and the topology is the optimization. The authority edge arrives from CADRE (REG-07), the grant the registry records from its version 2.0, carrying the authority to fix a deployment topology, and 2.5 still requires an institution to record the grant its own governance supplies. The grant is a relationship of authority, not an altitude, and the framework occupies none. DOI https://doi.org/10.6084/m9.figshare.33420352

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3. The Patterns (Normative)

Each pattern below is keyed to a clause disposition, not to a description of an institution. **The disposition selects the pattern; the institution does not.** The evidence every pattern requires is specified once, at Section 5, and is not restated per pattern.

3.1 Sovereign Silo

Applies when any clause the workload class engages carries the disposition `must-not-leave`.

Property. Complete duplication of the serving and storage path per jurisdiction. Nothing governed by the clause crosses the boundary, and the clause is honoured by the absence of a transfer event, not by a control over one. The cost is structural duplication. It does not fall with scale. An institution selecting this pattern commits to it for the life of the class. **This specification states no figure**, and 1.5 records why.

3.2 Federated

Applies when no clause carries `must-not-leave`, and at least one carries `may-move-derived-only`.

Property. Serving and data remain in jurisdiction; named derived artifacts move. The clause fixes which artifacts may move, and this framework fixes nothing about it.

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A selection of this pattern MUST name the derived artifacts moved, and each MUST be permitted by the governing clause. A derived artifact that carries the underlying data is not a derived artifact for this purpose, and an institution unable to establish that it does not carry the data has selected this pattern on an assumption rather than on a clause.

3.3 Regional Hub

Applies when every clause carries `may-transfer-under-mechanism` and the intersection of their accepted destinations is not empty.

Property. One jurisdiction serves the class, under the mechanisms the clauses name.

The selected hub MUST appear in the accepted destinations of every clause the class engages. A hub accepted by most clauses and not by all is not a hub, and this is the failure the intersection test exists to catch: it is easy to establish that transfer is permitted from each jurisdiction and never to check that all of them permit transfer to the same one.

The mechanism in force per inbound jurisdiction and the computed intersection MUST both be recorded with the assignment.

3.4 Hybrid is a composition, not a fourth alternative

Hybrid is the state of an estate whose workload classes resolve to more than one of the three patterns above. It MUST NOT be selected for a workload class.

The distinction carries the framework's falsifiability. A fourth alternative available at class level would be a catch-all. **A catch-all cannot be falsified.** Any workload the first three failed to serve would be assigned to it, and the enumeration would become unrefutable by construction.

Because Hybrid is a composition, it inherits the coverage of the three. It does not extend it. **The claim under test is that Sovereign Silo, Federated and Regional Hub exhaust the topologies available under a residency constraint at the level of a single workload class**, and 8.7 states the observation that would refute it.

The shared schema enforces this. Hybrid is absent from the enumeration a selection may assign to a class, and appears only as a derived value describing the composition across an estate.

Three is a claim. Four would be a container.

---

4. The Decision Instrument (Normative)

This section is the artifact registry entry REG-09 recorded as missing at the registry's version 1.0: without it, selecting a pattern required reading the source treatment, and the method could not be applied consistently by two architects working independently. Registry version 2.0 records that limitation as closed by this section [1].

4.1 Step one: take the boundary set

The architect MUST take the residency rule set produced by [4], at a stated version, and MUST NOT author or amend it. It carries, per jurisdiction, the clauses in force with their dispositions, exposures, primary sources and re-verification dates, and the examination scope of each authority involved.

4.2 Step two: classify the workloads

The architect MUST assign every workload in scope to a workload class, per 2.4. The classification MUST be recorded, and each class MUST state the clause identifiers it engages.

4.3 Step three: resolve

The pattern is determined by the most restrictive disposition among the clauses the class engages, in the order below. The procedure is normative and admits no discretion at this step.

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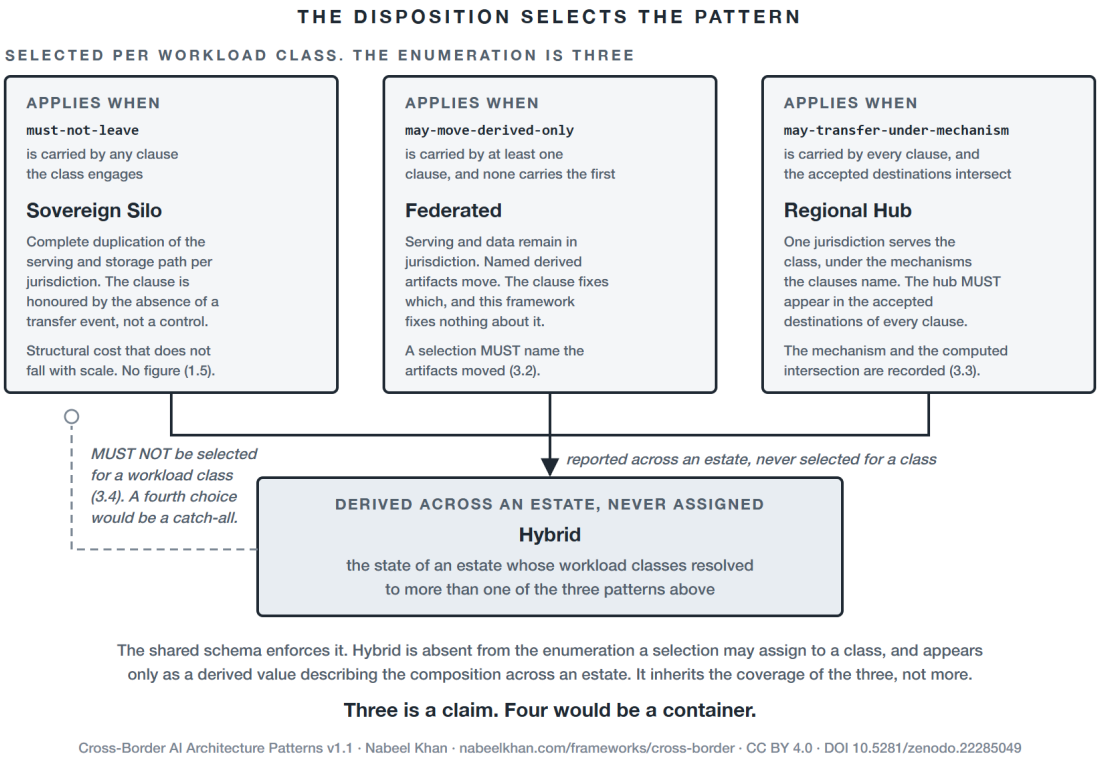


Figure 2. The three patterns and the composition. Each pattern is drawn against the clause disposition that selects it: `must-not-leave` selects Sovereign Silo, `may-move-derived-only` selects Federated, and `may-transfer-under-mechanism` with a non-empty destination intersection selects Regional Hub. Hybrid sits at estate level, drawn as the composition of classes that resolved differently rather than as a fourth box beside the three, because a fourth alternative at class level would be a catch-all and a catch-all cannot be falsified. DOI https://doi.org/10.6084/m9.figshare.33420187

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Condition on the clauses a workload class engages	Outcome
Any clause is `unresolved`, or is past its re-verification date	**Blocked.** No selection. Return to [4]
Any clause carries `must-not-leave`	**Sovereign Silo**
Otherwise, any clause carries `may-move-derived-only`	**Federated**, with the moved artifacts named per 3.2
Otherwise, every clause carries `may-transfer-under-mechanism` and the intersection of accepted destinations is non-empty	**Regional Hub**, at a destination in that intersection
Otherwise, every clause carries `may-transfer-under-mechanism` and the intersection is empty	**Blocked.** No accepted destination
The class engages no clause	**Out of scope**, per 1.4. Not a pattern
The pattern the rows above select cannot serve the class: with every clause correct and current, the workload cannot operate under that pattern's property at 3.1 to 3.3	**Blocked.** No pattern resolves. See 8.7

Rows one to six are exhaustive over the clause vocabulary and admit no discretion. The seventh does not. It is the one judgment in the instrument: the architect asserts that a pattern the rules selected cannot carry the workload. It MUST be recorded with the property at 3.1 to 3.3 the workload cannot satisfy, and it is the observation 8.7 turns on.

The order of the rows is normative. Reading them in any other order permits a class carrying both a `must-not-leave` clause and a transfer permission to resolve to Regional Hub, which is the most consequential error the instrument exists to prevent.

The governing clause MUST be recorded with the assignment. It is the single most restrictive clause the class engages, it is the clause that selected the pattern, and it MUST be recorded by identifier rather than by restatement.

4.4 Blocking states are outcomes, and MUST be recorded

A blocked class MUST be recorded with its reason and the clauses involved, and MUST NOT be assigned a pattern provisionally while the block is resolved.

The reason MUST be drawn from the closed vocabulary of the shared schema: `clause-unresolved`, `clause-expired`, `no-accepted-destination`, or `no-pattern-resolves`. The vocabulary is closed because a free-text reason permits a block to be recorded in terms that make it look like a scheduling matter, and three of these four are defects in the rule set while the fourth is a finding against this framework.

A workload class engaging no clause is out of scope rather than blocked. Recording it as blocked would overstate what the rule set constrains; assigning it a pattern would understate it.

A block is an outcome of the procedure.

4.5 Step four: record the declaration

Every selection MUST carry a BOE Declaration [2]: the boundary it honours, stated as the governing clauses; the optimizer it frees, stated as what the topology was chosen on once the clauses were fixed; and the evidence proving the boundary held, which is the routing evidence at Section 5.

4.6 What the instrument claims

Two architects given one rule set and one workload classification produce the same assignment, because every step above is determined by the rule set and none is left to judgment. That is the claim the instrument rests on and the claim the registry's `missing` limitation asked for, which registry version 2.0 records as closed by this section [1]. It has not been tested against two architects, which 8.4 records.

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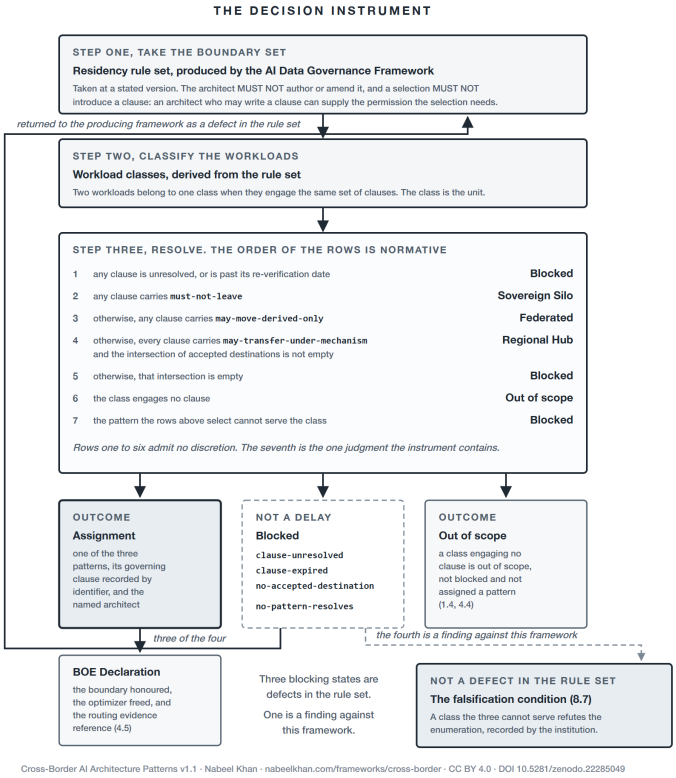


Figure 3. The decision instrument. The rule set enters from the AI Data Governance Framework and is not amended. Workloads are grouped into classes by the clause sets they engage; each class descends through the ordered resolution of 4.3 to a pattern, a blocking state or an out-of-scope record; and each assignment emits a BOE Declaration and a routing-evidence reference. The three blocking states that are defects in the rule set return upward to the producing framework; the fourth, no pattern resolves, points instead at this framework's own falsification condition. DOI https://doi.org/10.6084/m9.figshare.33420184

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5. Evidence (Normative)

An institution operating this framework MUST hold all of the following, and a framework whose evidence artifacts are not produced is not operating however completely it is documented.

1. **The rule set relied on**, at its version and issue date, held so that a past selection can be reconstructed against the clauses in force when it was made.
2. **The workload classification**, with the clause identifiers each class engages.
3. **The selection record**, one assignment or one blocking record per class, each naming the governing clause by identifier and the accountable enterprise architect.
4. **Residency attestations** per jurisdiction in which data is held or processed.
5. **Routing evidence**, sufficient to show per request that no clause was crossed.

A pattern selected without a documented boundary set is a topology choice rather than a compliance position [1]. The same holds for routing evidence. A topology that cannot show where a request was served has recorded an intention about traffic. It has not recorded a fact.

Accountable role. The named enterprise architect accountable for the deployment topology, occupiable by one person. Where the topology spans legal entities, the role is held in the entity operating the platform and the other entities are recorded as consulted [1].

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6. Conformance (Normative)

6.1 Self-declared

Conformance is **self-declared**. No conformity assessment body operates against this specification and no third party is accredited to assess a claim made under it.

6.2 What conformance requires

An institution claiming conformance for a stated estate asserts all of the following for the period claimed.

1. A rule set produced under [4] exists at a stated version, and no clause relied on was past its re-verification date at the time of selection.
2. Every workload in scope belongs to a recorded class stating the clauses it engages, per 4.2.
3. Every class carries an assignment, a blocking record or an out-of-scope record produced by the ordered resolution at 4.3, with its governing clause recorded by identifier.
4. No class is assigned Hybrid, per 3.4, and the selection introduces no clause, per 2.3.
5. Every assignment carries a BOE Declaration and a routing-evidence reference, per 4.5 and Section 5.
6. One named person is recorded as the accountable enterprise architect, and the authority under which that person fixes a topology is recorded, per 2.5.

6.3 What accompanies a claim

A published claim MUST carry the version claimed against, the date of the evidence, the estate covered, and the identifier and version of the rule set relied on.

6.4 What it does not mean

- It does not mean the institution complies with any residency law or transfer rule. The framework organizes a decision and its evidence; whether the decision satisfies an authority is the authority's determination.
- It does not mean the clauses in the rule set are correct. They are produced and sourced under [4], and their accuracy is that framework's requirement and the institution's obligation.
- It does not mean the institution has been assessed. Nobody assessed it.
- It does not mean a certification exists. None does.

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7. One Disagreement, Reported and Not Settled

Two published records in this corpus disagree about where this framework's output enters the deployment lifecycle, and this section reports it because the registry requires a disagreement between documents to be reported and not left standing [1].

The disagreement. The Five-Gate Deployment Model lists the deployment topology, where a system spans jurisdictions, as an entry criterion at **gate one** and again at **gate four**, in both cases supplied by this framework [3]. The registry's relationship register carries **one** edge from this framework to the deployment model, and its subject is the topology approved at **gate four** [1].

What this specification does about it: nothing, deliberately. The registry states that each dependency is asserted exactly once in a direction the register fixes, and that the registry governs placement where it and a member specification describe the same relationship [1]. **A deposit resolving this by writing the missing edge into itself would be authoring a registry edge**, which the register forbids. The disposition is the registry's, and this document does not anticipate it.

What an institution operating both today should do. Supply the topology at gate one as the deployment model requires, and record that it was supplied under that specification's entry criteria, not under a registry edge. Nothing is lost by doing so and the record stays accurate under either disposition.

This section reports only the disagreement bearing on this framework's own entry.

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8. Limitations, What Has Not Been Tested, and Falsification

8.1 No institution unconnected to the author has been observed operating the selection

There is no implementation profile, no scored instrument and no published result for the procedure as specified here. Everything in Sections 2 through 6 is architectural.

Nobody outside the author's practice has run it.

8.2 The authority grant is recorded from registry version 2.0

Version 1.0 of this specification recorded here that **the registry's relationship register recorded no authority grant to this framework**, while the registry's own account of the architecture holds that evidence cannot be produced without authority, and this framework produces evidence. It routed the disposition to the registry under Section 7's reasoning rather than working around it. **Registry version 2.0 resolved the gap at row 10 of its relationship register**, which grants this framework authority from CADRE (REG-07), held by the named enterprise architect, and records the resolution at its 4.9.12 and 6.9 [1]. The gap was in the register. The requirement at 2.5 now records a grant the register also records, rather than standing in for one it lacked.

8.3 The worked cases in the source treatment are composites

The cases in the source treatment [6] are disclosed there as composites. They illustrate a selection method. They are **not evidence that any named institution operates any pattern**. Nothing here rests on them.

8.4 The determinism claim is untested

4.6 claims that two architects given one rule set produce one assignment. That has not been tested with two architects. It is a property of the procedure as written, not an observation, and the difference matters: a procedure can be deterministic on paper and admit judgment where a workload is assigned to a class, which is the step at 4.2 the instrument constrains least.

8.5 The clauses of any rule set age, and this framework carries none

This specification states no residency rule, which removes the ageing problem from the document and not from practice. **Every clause an institution relies on ages**, and a boundary set carried forward without re-verification is not a boundary set [1]. The instrument treats an expired clause as a blocking state for that reason.

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8.6 Single-author instrument, with a conflict

This specification has one author, has not been peer reviewed, and has been reviewed by no institution. The author operates a consulting practice applying it, which is a conflict of interest stated here rather than in a footnote.

8.7 What would falsify the framework

The framework is falsified if a real boundary set requires a per-workload assignment in which some workload class can be served by none of Sovereign Silo, Federated or Regional Hub.

This is the falsification condition the registry records for REG-09 [1], restated without weakening. **Hybrid is not a fourth alternative for this purpose.** It is defined at 3.4 as a composition of the other three taken per workload class, and it inherits their coverage without extending it. Stating the condition against Hybrid's composition, and not against the four as a set, is deliberate, because a catch-all cannot be falsified.

The observation is recordable. A class in that position is recorded under the blocking reason `no-pattern-resolves`, which exists in the shared schema for no other purpose, and an institution producing one has produced the refutation without needing the author's agreement that it counts.

Two narrower findings would refute parts of it. If the ordered resolution at 4.3 produces an assignment an institution's authority rejects while the clauses relied on were correct and current, the resolution order is wrong. And if two architects given one rule set produce different assignments, 4.6 fails and the instrument has not closed the gap the registry recorded.

8.8 Research questions

1. Do two independent architects given one rule set and one workload classification produce the same assignment?
2. Does any real boundary set contain a workload class that none of the three patterns can serve?
3. What authority does an institution in practice name as granting an enterprise architect the power to fix a deployment topology, and does it resemble the grant the registry records from CADRE [1]?

Institutions and researchers willing to engage with any of these are invited to make contact through the ORCID record in the front matter. Question two would settle the framework, and it cannot be answered by its author.

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9. Versioning

The major number changes when a change could make a previously conformant institution non-conformant: a requirement in Section 6 is added or tightened, a pattern is added or removed, the resolution order at 4.3 changes, or the blocking vocabulary at 4.4 changes. **The minor number** changes for additions and clarifications that cannot.

A change to the shared schema is a change to the specification that publishes it [4]; this one tracks that schema's version rather than versioning it. Where this specification and registry entry REG-09 describe the same relationship, **the registry governs** [1], and a disagreement MUST be reported rather than settled here, per Section 7.

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10. References

Every reference to a work outside this corpus was verified against a live primary source, each with a deliberate failing control, because a check that cannot fail proves nothing. The date of verification is recorded at Appendix B.

[1] Khan, N. A. (2026). *The Defensible AI Framework Registry: Canonical Names, Definitions and Relationships for the Governed Production AI Discipline. Registry Specification.* Zenodo. Concept DOI https://doi.org/10.5281/zenodo.22170112

doi.org/10.5281/zenodo.22285049

[2] Khan, N. A. (2026). *A Pattern Language for Production LLM Platforms: Governed Routing, Agent Orchestration, and AI-Native Delivery. Pattern Language Specification.* Zenodo. Concept DOI https://doi.org/10.5281/zenodo.22109864

[3] Khan, N. A. (2026). *The Five-Gate Deployment Model: A Deployment Discipline for AI Systems. Deployment Model Specification.* Zenodo. Concept DOI https://doi.org/10.5281/zenodo.22170122

[4] Khan, N. A. (2026). *The AI Data Governance Framework: A Five-Stage Control System for the Data Boundary in AI Systems. Framework Specification.* Zenodo. Concept DOI https://doi.org/10.5281/zenodo.22285047

[5] Khan, N. A. (2026). *The MESA Framework: A Four-Altitude Diagnostic Model for Institutional AI Governance. Framework Specification.* Zenodo. Concept DOI https://doi.org/10.5281/zenodo.22109836

[6] Khan, N. A. (2026). *AI Governance and Compliance Frameworks for the Middle East: The Enterprise Playbook.* First Edition. iSystematic Inc. ISBN 978-1-0678960-1-0

A note on what is not cited. No statute, regulation, supervisory instrument or jurisdictional transfer rule is cited in support of any proposition here, because this document advances no proposition about what any of them requires. Clauses are the institution's to source, date and re-verify under [4].

Appendix A. Naming Crosswalk: The Three Patterns of the Source Treatment

Recorded so that work published under the former treatment remains citable. **A former name is not an error in the work that used it.**

A.1 The pattern names.

Source treatment, chapter 9 of [6]	This specification	Note
The Sovereign Silos Pattern	Sovereign Silo (3.1)	Singular, following the canonical form fixed by the registry [1]. The former plural remains valid for citation of the earlier work
The Federated Pattern	Federated (3.2)	Unchanged. The named-artifact requirement at 3.2 is new here
The Regional Hub Pattern	Regional Hub (3.3)	Unchanged. The destination-intersection test at 3.3 is new here
Not present	Hybrid (3.4)	**The source treatment specifies three patterns and does not name a fourth.** The nearest prior treatment is its account of institutions operating several patterns concurrently, described there as a pattern portfolio, which is a description of practice rather than a named pattern

A.2 Three claims of the source treatment that this specification does not carry.

The altitude. The source treatment states that pattern selection is a strategic decision at the second MESA level. **The registry repositions this framework outside the MESA altitudes entirely**, under the Boundary Invariant on the data residency boundary class [1], and 2.2 is written to the registry, which governs.

The count. The source treatment enumerates three patterns; this specification enumerates three assignable patterns and one composition, and 3.4 states why the composition is not a fourth alternative. **The change is substantive**, and the falsification condition at 8.7 depends on it.

The named jurisdictions, mechanisms and hub candidates. The source treatment names jurisdictions, transfer mechanisms and candidate hub locations, and dates from its own publication. **This specification names none** and requires every clause to carry its own primary source and re-verification date under [4]. A reader holding the book should treat its jurisdictional content as an ageing assertion to be re-verified rather than as an input to a selection.

A.3 One divergence from the registry entry, recorded here and resolved at registry version 2.0.

At the registry's version 1.0, registry entry REG-09 stated its scope phrase at 4.9.3 as four deployment architectures for multi-jurisdiction AI, and its definition at 4.9.5 said the patterns specify four architectures [1]. This specification says something narrower and more testable. **Three patterns are selectable at the level of a workload class, and 3.4 forbids selecting Hybrid for one**, because Hybrid is the state of an estate whose workload classes resolve to more than one of the three.

The disagreement is resolved by 11.3 of the registry [1], under which the specification governs the framework and the registry governs the framework's place in the architecture. How many patterns an architect may assign is a fact about the framework. **This document therefore governs the count, and the registry reconciled its entry with it at version 2.0**: its scope phrase at 4.9.3 now reads three deployment architectures for multi-jurisdiction AI, and the composition an estate reports when its workload classes resolve to more than one of them [1]. The registry's definition at 4.9.5 now says three [1].

A reader who finds four in the registry at its version 1.0, and three plus a composition here, follows this document on the count, as registry version 2.0 now does. Three things do not change with it: the framework name, Cross-Border AI Architecture Patterns; the identifier REG-09, which is normative and stable across any rename or restatement; and the four-value clause disposition vocabulary, which is a different enumeration and is unaffected.

Appendix B. Publication Provenance

B.1 What this document is. The first standalone specification of the Cross-Border AI Architecture Patterns, not peer reviewed. On deposit it receives a DOI, a timestamp and citability. **Deposit confers no standard status, and nothing in this document uses the word standard for itself.** No preprint identifier exists and no preprint relation is asserted in its metadata.

B.2 Relationship to prior publication. The patterns were published in chapter 9 of [6] as three architectures with a selection discipline stated in prose. This document is the first statement of the selection as a deterministic instrument, the first statement of Hybrid as a composition, and the first in which the position under the Boundary Invariant rather than under an altitude is carried into the normative sections.

B.3 What this deposit closes. At the registry's version 1.0, registry entry REG-09 recorded a `missing` limitation: no decision instrument existed, so the method could not be applied consistently by two architects working independently [1]. Section 4 is that instrument, and registry version 2.0 records the limitation as closed by it [1]. The schema it operates on is published with [4] rather than here, for the reason that specification gives at its 8.1.

B.4 Publication order. Published together with [4], whose schema it consumes and does not republish, and after the registry [1], the pattern language [2], the Five-Gate Deployment Model [3] and the MESA Framework [5], all cited at their concept DOIs. Registry entry REG-09 remains the canonical statement of this framework's name and relationships.

B.5 Verification method and date. Every reference in Section 10 was checked on **15 September 2026**. **This specification cites no work outside the corpus**, so that check has no external subject; every corpus reference was resolved through the Zenodo API, with a deliberately invalid record identifier confirmed not to resolve. The disagreement at Section 7 was established by reading both published records rather than by recollection, and that section names where each statement appears.

B.6 Funding and conflicts. The work is unfunded. The author operates a consulting practice applying this framework, a conflict of interest stated here and at 8.6.

Appendix C. Figure Specifications

Figures are deposited separately and carry their own identifiers. **Each caption in the body carries the argument on its own**, so that a reader who never sees a figure loses illustration rather than content.

Figure 1, the framework's position. This framework under the Boundary Invariant, with the four MESA altitudes drawn alongside and no line connecting them to it. One inbound edge carrying the rule set, two outbound edges carrying the topology and the routing evidence, and one inbound edge carrying the authority grant from CADRE

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(REG-07), drawn as a grant of authority rather than as an altitude. It MUST NOT place the framework at an altitude.

Figure 2, the three patterns and the composition. Three patterns, each keyed to the clause disposition that selects it, with Hybrid drawn at estate level rather than as a fourth box beside the three. The visual difference between the two levels MUST be unmistakable, because the falsification argument rests on it.

Figure 3, the decision instrument. The rule set entering unamended, workloads grouped into classes by clause set, the ordered resolution of 4.3 as one descending path, and three terminal kinds: a pattern with its declaration, a blocking state, and an out-of-scope record. The three blocking states that are rule-set defects MUST be drawn returning to the producing framework; `no-pattern-resolves` MUST be drawn pointing at the falsification condition instead.

Document History

Version	Date	Summary
1.0	15 September 2026	First public version. Three assignable patterns and one composition, the deterministic decision instrument with its blocking states, the recorded absence of an authority grant, the disagreement reported to the registry at Section 7, and the naming crosswalk to the three-pattern source treatment
1.1	19 October 2026	The authority grant aligned with registry version 2.0 [1], which grants this framework authority from CADRE (REG-07) at row 10 of its relationship register. 2.5 and 8.2 now record the grant rather than its absence, with the requirement at 2.5 unchanged; the abstract, 1.3, the third research question at 8.8, Figure 1, its caption and its specification at Appendix C, and the trademark notice follow. The statements recording the registry's `missing` limitation for this entry, at the abstract, 1.3, Section 4, 4.6 and B.3, now record it as closed by Section 4, and A.3 records the count as settled at three by registry version 2.0 at its 4.9.3 and 4.9.5. The attribution footers of Figures 2 and 3 read version 1.1. A minor version under Section 9, because registry version 2.0 already holds each of these and this change corrects the specification to it rather than tightening a requirement

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About the Author

Nabeel Khan is an enterprise AI architect and governance advisor, and Founder & Solutions Architect at iSystematic Inc. He has spent twenty-five years architecting enterprise data and machine-learning systems in government, healthcare, financial services and telecommunications across three continents, grounded in TOGAF, DMBOK, ISO 27001 and SOC 2.

He works on governed production AI; systems whose authority is bounded by the architecture, whose decisions carry the evidence needed to defend them, and whose behaviour the institution can explain afterwards. The frameworks specified here, and the regulatory knowledge behind them, are published as machine-accessible AI expertise and are callable over MCP without a key at mcp.nabeelkhan.com/api/mcp, so a reader may query this specification's contents directly and receive answers carrying their source.

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The schema this specification consumes, `residency-rule-set-schema-v1.0.json`, is published under the same licence with [4].

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